
Multiplierless 2D Output Transform Unit (output_transform_2d.sv)
Accumulated $M4 \times 4$ INT36 (Signed)
16 × 36b
4 × 1D Col Transformsoutput_transform_1d $P = A^T \cdot M_{p_0} = m_0 + m_1 + m_2 p_1 = m_1 - m_2 - m_3$ Dim: $4 \times 4 \to 2 \times 4$ (38b)
8 × 38b
REG STAGE 1 (38b)
Transpose P^T Bus rewiring $2 \times 4 \to 4 \times 2$
8 × 38b
2 × 1D Row Transformsoutput_transform_1d $Y' = A^T \cdot P^T{}_{y'_0} = p'_0 + p'_1 + p'_2 y'_1 = p'_1 - p'_2 - p'_3$ Dim: $4 \times 2 \to 2 \times 2$ (40b)
4 × 40b
Scaling & Saturation Arithmetic Right Shift:>>> 6 ($1/64$ scale factor) Clamp to Signed INT16:$[-32768, +32767]$
4 × 16b
REG STAGE 2 (16b)
4 × 16b
Output Tile $Y 2 \times 2$ INT16out_valid